

Project Plan: Special Olympics Data Case

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1. Project Objective

The goal of this project is to transform raw ERP data from Special Olympics into an actionable Power BI dashboard. The solution will utilize a Python-based ETL pipeline following OOP principles, a local MySQL database for storage, and a professional semantic model in Power BI to answer key business use cases regarding athlete performance and regional growth.

2. Weekly Breakdown

Week	Phase	Key Tasks & Activities	Deliverables
Week 1 (May 1 – May 7)	Foundation & Modeling	Analyze historical XLSX files (2015-2025). Design a normalized Star Schema. Set up GitHub repository and local MySQL environment.	<u>r0975016_ProjectPlan.pdf</u> <u>r0975016_Model.pdf</u>
Week 2 (May 8 – May 14)	Python ETL & Data Pipeline	Develop Python scripts (main.py) using OOP. Implement Medallion Architecture (Bronze to Gold layers). Ingest cleaned data into MySQL.	<u>r0975016_Processed_Data.zip</u> <u>r0975016_Repo_Archive.zip</u>
Week 3 (May 15 – May 20)	Power BI & Documentation	Create report wireframes. Connect Power BI to MySQL. Develop DAX measures and interactive visuals. Final data validation and technical documentation.	<u>r0975016_Wireframe.pdf</u> <u>r0975016_Dashboard.pbix</u> <u>r0975016_Documentation.pdf</u>

3. Technical Approach

- Version Control: Frequent commits to GitHub to document progress and logic.
- Data Integrity: Addressing inconsistencies across historical data files (e.g., missing columns in 2023) through Python-based cleaning.
- Bonus Features: Full integration with MySQL and implementation of the Medallion Architecture for advanced data grading.

4. Risk Management

- Tight Deadline: Mitigated by a dedicated 2-3 hour daily work block.
- Validation: Ensuring dashboard totals match raw Excel sources through a final audit phase.